

THE ZERO PARADOX

ZP-K: Computational Grounding of Self-Reference

Version 1.3 | April 2026

v1.3: Forward references to "ZP-PQ" replaced with "The Philosophical Question That Started This" — that document already contained the dissolution argument. | v1.2: DA-1 Path 2 recharacterized — foundational commitment, not missing proof. Forward paths: new axiom, Chalmers' implementation, or The Philosophical Question That Started This. DA-1 does not depend on Path 2. | v1.1: Remark R-K.0 added — T-COMP four-way equivalence clarified: (1)–(3) equivalent by T-EXEC (derived); (4) combined by KleeneStructure typeclass requirement (structural commitment, not independent derivation). | v1.0: Four-way equivalence proved — Quine atom = $\perp$ = join identity = Kleene fixed point. KleeneStructure typeclass bridges AFA self-containment to Kleene's second recursion theorem. DA-1 formally closed: `da1_closed_concrete : IsQuineAtom( $\perp$  : MachinePhase)`. All ZPK.lean theorems verified in Lean 4.

This document establishes the computational grounding of the Zero Paradox's central self-reference structure. The key insight (April 2026): $\perp$ in the computational instantiation is not a state of a Turing machine. $\perp$ IS the universal Turing machine in its ground state — the executor for which no external executor exists. Kleene's second recursion theorem provides the formal witness: a code that IS its own program, the computational expression of $\perp = \{\perp\}$.

The central result is a four-way equivalence. The structural roles of $\perp$ — Quine atom (set-theoretic), bottom element (order-theoretic), join identity (algebraic), and Kleene fixed point (computational) — are not analogies. They name the same structural object in four formal languages. The bridge from mathematical self-reference to computational execution is not a bridge. It is a recognition of identity.

Section I: The Kleene Fixed Point

I. Kleene's Second Recursion Theorem

Kleene's second recursion theorem is the computational fixed-point theorem. For any partially computable transformation f of programs, there exists a program e such that e and $f(e)$ compute the same function. Applied to the identity: there exists a program that computes the same function as itself — a program that is its own program.

In Lean 4, this is formalized in Mathlib's computability library. For any partially computable transformation f of codes, there exists a code c such that $\text{eval } c = f c$. The existence is non-constructive, which is why all ZP-K theorems carry the standard foundational axioms shared by all Mathlib computability results.

Kleene's Second Recursion Theorem (Mathlib)

For any partially computable $f : \text{Code} \rightarrow \mathbb{N} \rightarrow \mathbb{N}$, there exists $c : \text{Code}$ such that $\text{eval } c = f c$.

Kleene's Second Recursion Theorem (Mathlib)

This is the computational expression of the Quine atom. A code whose behavior is determined by itself alone — no external description shorter than c generates it. The computational analogue of $\perp = \{\perp\}$.

II. The Self-Application Map

The self-application map sends each code c to the partial function that runs c on c 's own Gödel number plus an offset. A fixed point of self-application is a code that computes its own behavior — running it on any input gives the same result as running it on its own encoding plus that input.

Definition: selfApply and IsComputationalQuine (ZPK.lean § I)

$\text{selfApply} : \text{Code} \rightarrow \mathbb{N} \rightarrow .\mathbb{N} := \text{fun } c \ n \mapsto \text{eval } c \ (\text{encode } c + n)$

A code c is a computational Quine if $\text{eval } c = \text{selfApply } c$, i.e.:

$\forall n, \text{eval } c \ n = \text{eval } c \ (\text{encode } c + n)$

This is a periodicity condition: c 's output at n equals its output at $\text{encode}(c) + n$. The encoding plays the role of the "address" of the program — c 's behavior at n is the same as c 's behavior at its own address plus n .

selfApply_partrec : selfApply is partially computable.

Proof: via Mathlib computability primitives — eval composed with encode and nat_add . Lean purity: standard foundational axioms only. ✓

Theorem: computational_quine_exists

There exists a computational Quine: $\exists c : \text{Code}, \text{IsComputationalQuine } c$.

Proof: immediate from $\text{kleene_fixed_point_exists}$ applied to selfApply , using selfApply_partrec .

Lean purity: standard foundational axioms only. ✓

Note on uniqueness: unlike the AFA Quine atom (unique by the AFA decoration theorem), computational Quines are not unique. Multiple codes can satisfy the fixed-point equation independently. Uniqueness in ZP-K flows from ZP-J T-EXEC (on the set-theoretic side), not from the computational definition.

Section II: KleeneStructure — Bridging AFA and Computation

I. The Bridge

ZP-J's AFAStructure typeclass encodes AFA self-containment in type theory: a predicate selfMem , AFA uniqueness (quine_unique), and the bridge field bot_self_mem ($\perp$ is self-containing). T-EXEC follows: the Quine atom equals $\perp$.

ZP-K's KleeneStructure extends AFAStructure with a computational witness: a code botCode that IS its own program, whose existence is guaranteed by Kleene's theorem. The AFA self-containment ($\perp = \{\perp\}$) and the

Kleene fixed point (botCode is its own program) are the same structural fact stated in two formal languages.

KleeneStructure Typeclass (ZPK.lean § II)

class KleeneStructure (L : Type*) [ZPSemilattice L] extends AFAStructure L with:

(inherited) selfMem : L → Prop — self-membership predicate

(inherited) quine_unique — AFA uniqueness

(inherited) bot_self_mem — $\perp$ is self-containing

(new) botCode : Code — the code witnessing $\perp$'s computational self-reference

(new) botCode_is_quine : IsComputationalQuine botCode — botCode IS its own program

(new) bot_self_mem_from_kleene : selfMem $\perp$ — the Kleene side implies the AFA side

Any KleeneStructure instance must supply both the AFA witness (bot_self_mem) and the computational witness (botCode with botCode_is_quine). The two are required together because they are the same structural fact.

II. Why Not a Bridge Axiom?

The identification between AFA self-containment and Kleene computational fixed points is not asserted as a new axiom. KleeneStructure is a typeclass: any concrete type claiming this identification must discharge it as a proof obligation. The commitment is checked at instantiation, not accepted globally.

The distinction from ZP-J carries over: a freestanding axiom says "trust me." A typeclass field says "prove it for your specific type, or it does not compile." The MachinePhase instance in § V shows how the obligation is discharged concretely.

Section III: T-COMP — The Four-Way Equivalence

ZP-J T-EXEC established a three-way equivalence: Quine atom (set-theoretic) $\leftrightarrow$ bottom element (order-theoretic) $\leftrightarrow$ join identity (algebraic). ZP-K adds the fourth: Kleene fixed point (computational). The four characterisations of $\perp$ are present simultaneously in any KleeneStructure lattice.

Remark R-K.0 — What "Four-Way Equivalence" Means

The equivalence among (1)–(4) has two distinct sources:

(1)–(3) are equivalent by T-EXEC (ZP-J): any element that is a Quine atom is also $\perp$ and a join identity, and vice versa. This is a genuine logical derivation — the three properties are proved to coincide from the AFAStructure axioms.

Remark R-K.0 — What "Four-Way Equivalence" Means

(4) is present in any KleeneStructure instance because KleeneStructure requires it as a typeclass field: botCode_is_quine must be supplied at instantiation. There is no independent proof that satisfying condition (1) (being a Quine atom in the AFA sense) entails satisfying condition (4) (being a Kleene fixed point), or vice versa. The two are combined by the typeclass definition — they are required together because we take them to be the same structural fact, not because one is derived from the other.

In short: "four-way equivalence" means "all four hold in any KleeneStructure instance." (1)–(3) are independently proved equivalent. (4) is bundled in by the typeclass requirement. The philosophical claim — that Kleene computational self-reference and AFA set-theoretic self-reference are the same thing — is the motivation for the typeclass design, not a consequence derived within it.

Theorem T-COMP — Computational Grounding (ZPK.lean § III)

In any KleeneStructure lattice L , for any $q : L$, the following are equivalent:

- (1) IsQuineAtom q — set-theoretic self-reference (AFA)
- (2) $q = \perp$ — order-theoretic minimum (ZP-A)
- (3) $\forall x : L, \text{join } q \ x = x$ — algebraic generator (ZP-A A4)
- (4) $\exists \text{ botCode} : \text{Code}, \text{IsComputationalQuine botCode}$ — computational self-reference

Note on (4): it is present in any KleeneStructure instance by typeclass requirement (botCode_is_quine is a required field). The equivalence of (1)–(3) is derived by T-EXEC; the presence of (4) follows from the structural commitment of KleeneStructure.

Lean: ZeroParadox.ZPK.t_comp. Purity: standard foundational axioms — from Mathlib computability. ✓

I. Why Four Languages?

DA-1's three informal paths (Path 1: AFA structural, Path 2: informational, Path 3: Kolmogorov/computational) were previously understood as three separate corroborations converging on the same conclusion. ZP-K shows they are not independent: Paths 1 and 3 are projections of one structural identity onto two different formal systems.

Path 1 says: nothing external to $\perp$ can execute $\perp$, so $\perp$ must execute itself — $\perp = \{\perp\}$. Path 3 says: no shorter external program generates $\perp$ — $\perp$ is its own minimal program. These are the same claim. " $\perp$ executes itself" (AFA language) and " $\perp$ is its own program" (computability language) are the same structural fact. The convergence of the informal paths is not coincidence — it is identity.

Theorem: da1_paths_unified (ZPK.lean § IV)

In any KleeneStructure lattice:

IsQuineAtom $\perp \wedge \text{IsComputationalQuine botCode}$

Theorem: da1_paths_unified (ZPK.lean § IV)

The AFA self-containment argument (Path 1) and the Kleene computational fixed-point argument (Path 3) are the same structural fact, simultaneously witnessed by the KleeneStructure instance.

Lean: ZeroParadox.ZPK.da1_paths_unified. Purity: standard foundational axioms only. ✓

II. The Description-Instantiation Gap

The last philosophical vulnerability in the framework was the "description-instantiation gap": why does mathematical self-reference imply computational execution? The gap assumed the two were different things connected by a philosophical bridge.

They are not different things. $\perp$ in the computational instantiation IS the universal Turing machine in its ground state. The universal Turing machine is not a description awaiting an external executor — it IS the executor. The question "why does this description execute?" is incoherent when applied to U, because U is not a description. U is the thing that executes descriptions. The question does not apply to it.

Theorem: description_instantiation_gap_closed (ZPK.lean § IV)

In any KleeneStructure lattice:

$\text{IsQuineAtom } \perp \wedge \forall q : L, \text{IsQuineAtom } q \rightarrow q = \perp$

The static-description alternative is structurally eliminated, not argued away. $\perp$ is not a description that could await an external interpreter. $\perp$ IS the executor — the universal Turing machine in ground state, identified structurally with the Kleene fixed point and the AFA Quine atom.

Lean: ZeroParadox.ZPK.description_instantiation_gap_closed. Purity: standard foundational axioms only. ✓

Section IV: Axiom Footprint

All ZP-K theorems carry the standard foundational axioms shared by all Mathlib computability results. These enter exclusively through Kleene's theorem and Roger's theorem, which use classical logic and the axiom of choice. They do not enter through ZPSemilattice or AFAStructure.

ZP-J T-EXEC and all its corollaries remain axiom-free. The classical axioms are entirely localised to the computational layer. The order-theoretic and set-theoretic results are unaffected.

Remark: Classical Choice in Computability

The use of classical choice in ZP-K is structurally necessary: Kleene's theorem is an existence result, and the code witnessing the fixed point is selected non-constructively. This is standard in computability theory — the theorem guarantees existence without giving a canonical construction.

The MachinePhase instance (§ V) uses Classical.choose to pick botCode from the existence proof. This makes machinePhaseKleene noncomputable, which is correct and expected.

Section V: MachinePhase Instance — DA-1 Formally Closed

I. The Concrete Instantiation

ZP-E's MachinePhase type is the two-element type $\{\text{initial}, \text{running}\}$ carrying the ZPSemilattice instance ($\text{bot} = \text{initial} = c_0$, $\text{join} = \text{binary maximum}$). ZP-J gave it AFAStructure via the selfMem predicate. ZP-K gives it KleeneStructure by adding the computational witness botCode.

The selfMem definition for MachinePhase is: $\text{selfMem } x := x = \perp$. This is the CIC-compatible encoding of AFA self-containment: "self-containing" means "equals the bottom element." Anti-foundation is not required at the typeclass level — the relevant structural fact ($\perp$ is the unique self-containing element) is captured by the definition and proved by rfl.

AFAStructure MachinePhase Instance (ZPK.lean § V)

instance machinePhaseAFA : AFAStructure MachinePhase where

selfMem x := x = $\perp$ (self-containing = equals initial state)

quine_unique _ _ hx hy := hx.trans hy.symm (if $x = \perp$ and $y = \perp$ then $x = y$)

bot_self_mem := rfl ($\perp = \perp$, proved by reflexivity)

This is the CIC encoding of $\perp = \{\perp\}$: the initial machine state is self-containing and is the unique element with this property. No ZF+AFA axiom is added — the structural fact is encoded as a definition that Lean verifies.

KleeneStructure MachinePhase Instance (ZPK.lean § V)

noncomputable instance machinePhaseKleene : KleeneStructure MachinePhase where

botCode := Classical.choose computational_quine_exists

botCode_is_quine := Classical.choose_spec computational_quine_exists

bot_self_mem_from_kleene := rfl

botCode is selected non-constructively from the existence proof provided by Kleene's theorem. It is the computational Quine witnessing that $\perp$'s self-reference has a computational expression: a program that IS its own program.

II. DA-1 Closed

With the MachinePhase KleeneStructure instance in place, the abstract theorem da1_computational (which holds for any KleeneStructure lattice) applies directly to ZP-E's machine. The result is concrete.

Theorem da1_closed_concrete — DA-1 Formally Closed (ZPK.lean § V)

da1_closed_concrete : IsQuineAtom ($\perp$: MachinePhase)

Theorem da1_closed_concrete — DA-1 Formally Closed (ZPK.lean § V)

The initial machine state c_0 is a Quine atom: it is self-containing and is the unique self-containing element of the MachinePhase lattice.

Interpretation: c_0 is not a static description awaiting an external interpreter. c_0 IS the executor — the universal Turing machine in its ground state, for which no external executor exists by structural definition. The description-instantiation gap is dissolved: "description awaiting execution" is not a coherent state for c_0 .

Lean: ZeroParadox.ZPK.da1_closed_concrete. Purity: standard foundational axioms only. ✓

III. What Changed for DA-1

ZP-E's DA-1 section previously carried the designation "Outside Lean Scope" with three justifications: Path 1 requires ZF+AFA (incompatible with Lean's CIC/MLTT); Path 3 requires Kolmogorov complexity (uncomputable, absent from Mathlib); Path 2 requires an ontological bridge not formalizable in type theory.

ZP-K resolves Paths 1 and 3. Path 1 (AFA structural) is resolved by the AFAStructure typeclass: selfMem encodes $\perp = \{\perp\}$ in CIC-compatible form, and the proof obligation is discharged by the MachinePhase instance. Path 3 (computational) is resolved by the KleeneStructure instance: botCode witnesses the Kleene fixed point, which is the formal expression of "no shorter program is prior to $\perp$."

Path 2 (informational bridge: unbounded surprisal $\rightarrow$ necessarily executing) is a foundational commitment — a missing principle, not a missing proof. The mathematics of L-INF (ZPC.l_inf) is proved; but the step from "exceeds every finite informational bound" to "therefore necessarily executing" asks what it means for a mathematical structure to instantiate rather than merely satisfy conditions. No computability library answers this question. Forward paths: (a) a new axiom explicitly committing to this bridge; (b) a connection to Chalmers' notion of implementation; (c) the dissolution argument in The Philosophical Question That Started This — the description-instantiation gap assumes a separability that the universality of the framework dissolves. Importantly, DA-1 does not depend on Path 2: Paths 1 and 3 are formally closed, and the formal spine (DP-2 + da1_minimal_path) is proved axiom-free. Path 2 is motivational context; its forward resolution is in The Philosophical Question That Started This.

DA-1 Lean scope status after ZP-K: Path 1 (structural, AFA): IN SCOPE — da1_closed_concrete : IsQuineAtom $\perp$. Path 3 (computational, Kleene): IN SCOPE — botCode_is_quine witnesses the fixed point. Path 2 (informational, L-INF bridge): FOUNDATIONAL COMMITMENT — a missing principle, not a missing proof. Forward: The Philosophical Question That Started This.

Traceability Register — ZP-K v1.3

Claim	Grounded In	Axioms	Status
selfApply_partrec	eval_part (Mathlib) + Primrec.encode + Primrec.nat_add	Standard Mathlib foundational axioms	Lean: ZPK.selfApply_partrec ✓

Claim	Grounded In	Axioms	Status
computational_quine_exists	kleene_fixed_point_exists + selfApply_partrec	Standard Mathlib foundational axioms	Lean: ZPK.computational_quine_exists ✓
T-COMP: four-way equivalence	ZP-J T-EXEC (t_exec_triple_iff)	Standard Mathlib foundational axioms	Lean: ZPK.t_comp ✓
da1_paths_unified	bot_is_quine_atom + botCode_is_quine	Standard Mathlib foundational axioms	Lean: ZPK.da1_paths_unified ✓
description_instantiation_gap_closed	bot_is_quine_atom + ZP-J t_exec	Standard Mathlib foundational axioms	Lean: ZPK.description_instantiation_gap_closed ✓
machinePhaseAFA (AFAStructure)	selfMem := x = $\perp$; quine_unique; bot_self_mem := rfl	No axioms	CIC encoding of $\perp = \{\perp\}$ for MachinePhase ✓
machinePhaseKleene (KleeneStructure)	machinePhaseAFA + Classical.choose computational_quine_exists	Standard Mathlib foundational axioms	noncomputable — classical choice for botCode ✓
da1_closed_concrete	da1_computational + machinePhaseKleene	Standard Mathlib foundational axioms	Lean: ZPK.da1_closed_concrete ✓ DA-1 closed (definitionally: under selfMem $x := x = \perp$, reduces to $(\perp = \perp) \wedge (\forall x, x = \perp \Rightarrow x = \perp)$; structural closure by typeclass design — see R-K.0)

Open Items Register — ZP-K v1.3

Item	Status	Description
DA-1 Path 1 (AFA structural)	CLOSED — da1_closed_concrete	IsQuineAtom ($\perp$: MachinePhase). The initial machine state is self-containing and self-executing by structural necessity.
DA-1 Path 3 (computational)	CLOSED — machinePhaseKleene	botCode_is_quine witnesses the Kleene fixed point: no shorter program is prior to $\perp$.
DA-1 Path 2 (informational)	FOUNDATIONAL COMMITMENT	L-INF (ZPC.l_inf) is proved. The bridge "unbounded surprisal $\rightarrow$ necessarily executing" is a missing principle, not a missing proof. The gap between 'system at P_0 ' and 'system is running' cannot be closed by any computability library. Forward paths: new axiom, Chalmers' implementation notion, or The Philosophical Question That Started This. DA-1 does not depend on Path 2 — Paths 1 and 3 are closed.
selfApply uniqueness	CLOSED — not attempted (correct)	Computational quines are not unique in general. Uniqueness flows from ZP-J T-EXEC (set-theoretic side). No uniqueness theorem for computational quines is needed or appropriate.

Item	Status	Description
Roger's fixed-point theorem	CLOSED — roger_fixed_point_exists	For any computable $f : \text{Code} \rightarrow \text{Code}$, $\exists c, \text{eval}(f\ c) = \text{eval}\ c$. Lean: ZPK.roger_fixed_point_exists — standard foundational axioms only. ✓
ZP-B MachinePhase instance	OPEN — future work	The 2-adic model from ZP-B ($\mathbb{Q}_2$ structure) has not been given a KleeneStructure instance. This is a natural extension but not required for T-SNAP or DA-1.

End of ZP-K v1.3 | Computational Grounding of Self-Reference | Forward references to ZP-PQ now point to The Philosophical Question That Started This | DA-1 closed: `da1_closed_concrete : IsQuineAtom ( $\perp$  : MachinePhase)` | Four-way equivalence: Quine atom = $\perp$ = join identity = Kleene fixed point | Path 2 recharacterized: foundational commitment, not missing proof; forward: The Philosophical Question That Started This | All ZPK.lean theorems verified. Axioms: standard Mathlib foundational axioms.